

1. IIT-JEE Syllabus

Various concepts of Acids and Bases, Relative Strengths of Inorganic and Organic Acids and Bases.

2. Introduction

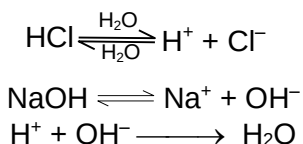
There are several so-called theories of acids and bases, but they are not really theories but merely different definitions of what we choose to call an acid or a base. Since it is only a matter of definition, no theory is more right or wrong than any other, and we use the most convenient theory for a particular chemical situation. So before we talk of strength of acids and bases, we need to know several theories.

3. Modern Concepts of Acids and Bases

Following are the important modern concept of acids and bases:

3.1 Arrhenius Concept - The Water Ion System

According to this concept, an acid is any hydrogen containing compound which gives H^+ ions in aqueous solution and a base which gives OH^- ions in aqueous solution. The HCl is an acid and NaOH is a base and the neutralisation process can be represented by a reaction involving the combination of H^+ and OH^- ions to form H_2O .



Utility

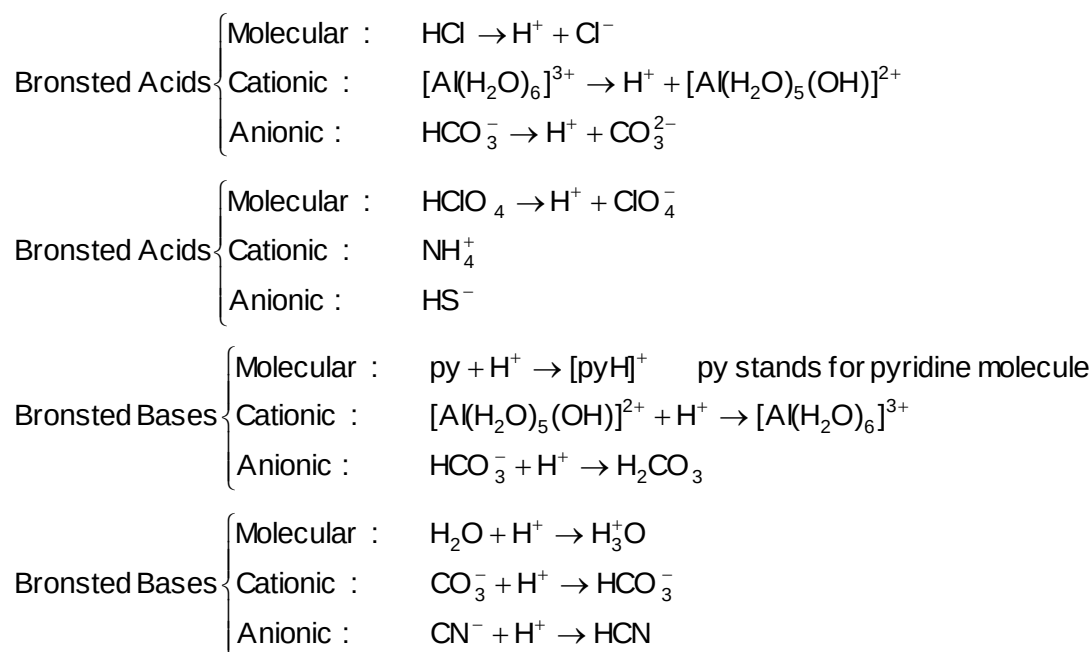
- Since the reaction representing neutralisation process involves the combination of H^+ and OH^- ions, the approximately constant molar heat of neutralisation would be expected. Thus the constant heat of neutralisation of a strong acid by a strong base is readily understandable in terms of this concept.
- This concept has offered a means of correlating catalytic behaviour with the concentration of the H^+ ion.

Limitations

- The acid or base property of a substance was not inherent in it, but was dependent on the presence of water. According to this concept, HCl is regarded as an acid only when dissolved in H_2O and not in some other solvent such as C_6H_6 or when it exists in the gaseous form.
- The neutralisation process is limited to those reactions which can occur in aqueous solution only, although reactions involving salt formation do occur in many other solvents and even in the absence of solvents.
- It cannot explain the acidic character of certain salts such as $AlCl_3$ in aqueous solution.

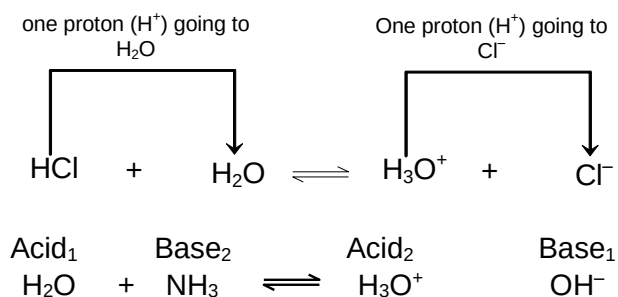
3.2 Bronsted - Lowry Theory - The Proton - donor - Acceptor System

Bronsted and Lowry in 1923 independently proposed a more general definition of acids and bases. According to them, an **acid** is defined as any hydrogen containing material (a molecule or a cation or an anion) that can release a proton (H^+) to any other substance, whereas a **base** is any substance (a molecule or a cation or an anion) that can accept a proton from any other substance. In short, an acid is a proton -donor and a base is a proton - acceptor.

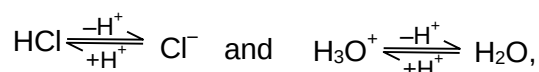


Conjugate Acid - Base Pairs

Consider a reaction

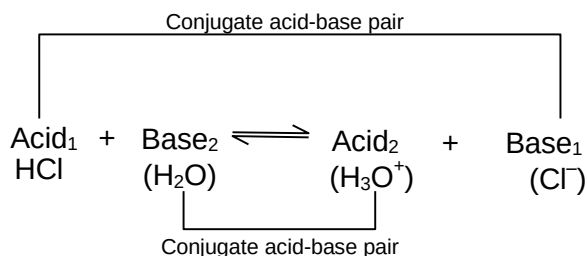


In this reaction HCl donates a proton to H_2O and is, therefore, an acid. Water, on the other hand, accepts a proton from HCl , and is, therefore, a base. In the reverse reaction which at equilibrium proceeds at the same rate as the forward reaction, the H_3O^+ ions donates a proton to Cl^- ion, hence H_3O^+ ion is an acid. Cl^- ion, because it accepts a proton from H_3O^+ ion, is a base. Acid base pairs such as.



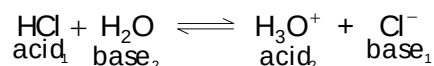
the members of which can be formed from each other mutually by the gain or loss of proton are called *conjugate acid - base pairs*.

If in the above reaction, the acid HCl is labelled Acid₁ and its conjugate base viz. Cl⁻ as Base₁ and further, if H₂O is designated Base₂ and its conjugate acid viz. H₃O⁺ as Acid₂, the equilibrium can be represented by a general equation.



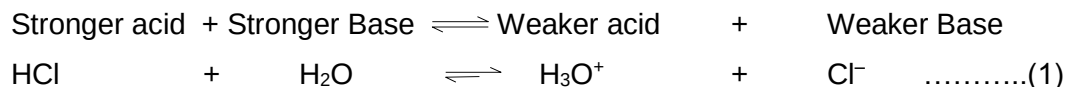
This is the fundamental equation representing the relationship between an acid and a base on the basis of Bronsted concept. Thus on the basis of this concept Acid₁ and Base₁ form one conjugate acid-base pair and Acid₂ and Base₂ form another conjugate acid-base pair.

Two important axioms of the Bronsted concept and position of equilibrium in acid-base reactions:



In the equilibrium mixture two acid HCl and H₃O⁺ ion are competing to donate protons to a base. Since HCl wins, it is the stronger acid. Similarly two bases, H₂O and Cl⁻ ion, are competing to accept protons. Here H₂O is the stronger base. It will be seen that the stronger acid, HCl, has the weaker conjugate base Cl⁻ ion and the stronger base, H₂O, has weaker conjugate acid, H₃O⁺ ion. *The stronger acid and weaker base form one conjugate acid - base pair and weaker acid and stronger base form another conjugate acid base pair.* It is quite evident that HClO₄ is the strongest acid; its conjugate base ClO₄⁻ ion, is consequently the weakest base. CH₄ and H₂ are the weakest acids; their conjugate bases, CH₃⁻ ion and H⁻ ion respectively, are consequently the strongest bases.

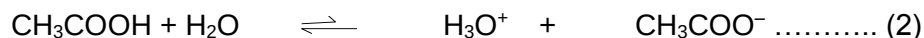
As a stronger acid, HCl is highly ionised even in concentrated aqueous solution. At equilibrium, the above reaction proceeds to the right, with most of HCl ionised to form H₃O⁺ and Cl⁻ ions. This fact can be illustrated by using arrows of unequal length to designate the forward and reverse reactions respectively. Thus.



The longer arrow indicates that the position of equilibrium lies to the right.

In the ionisation of CH₃COOH in H₂O, equilibrium is reached when the reaction has proceeded to the right only to slight extent, with only a small fraction of the CH₃COOH present in the form of ions.

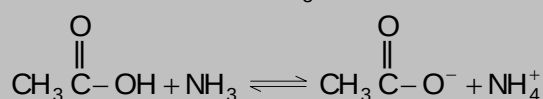
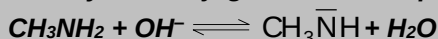
Weaker acid + Weaker base $\rightleftharpoons$ Stronger acid + Stronger base



Here the longer arrow indicates that the position of equilibrium lies to the left.

Evidently H_3O^+ ion in equilibrium (2) is a stronger acid and CH_3COO^- ion is a stronger base. It is also evident that the stronger acid H_3O^+ ion has the weaker conjugate base, H_2O and the stronger base, CH_3COO^- has the weaker conjugate acid, CH_3COOH . We thus see that all the proton transfer reactions (i.e., protolysis reactions) run downhill to form predominantly the weaker acid and the weaker base.

Exercise 1: Identify the conjugate acid base pair



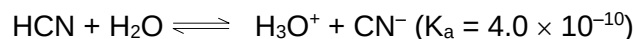
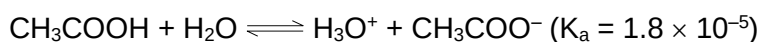
Relative Strengths of Acids and Bases

We have already discussed several definitions of acids and bases. It is interesting to study the general trends in acid or base strength among such compounds.

The protonic acids may be broadly classified as (i) hydroacids and (ii) oxo acids. In hydroacids, the acidic hydrogen atom is directly linked to some second element. Binary hydrides like HF, HCl, H_2S are such examples. In oxo acids, the acidic hydrogen is linked to some central atom via an oxygen atom e.g. for H_2SO_4 , HNO_3 or HClO_4 .

According to Bronsted concept, a stronger acid has a stronger tendency to donate a proton and a strong base has a strong tendency to accept a proton. At least two general methods are generally used for the comparison of relative acidity of given acids.

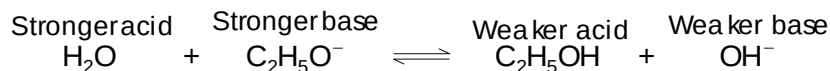
- i) The first of these consists of making a comparison of proton-donating tendencies of different acids towards the same base. For moderately strong acids, H_2O is generally used as the base. Suppose we have to compare the acidic strengths of CH_3COOH and HCN . Experimentally it has been observed that the ionisation or acidity constant, K_a for CH_3COOH and HCN at 25°C is 1.8×10^{-5} and 4.0×10^{-10} respectively.



CH_3COOH is, therefore, a stronger acid than HCN and CN^- ion is a stronger base than CH_3COO^- ion.

- ii) The second method is the competitive protolysis method. In this method one acid is added to the conjugate base of another and the equilibrium concentration are determined experimentally. For example, when NaOC_2H_5 is added to H_2O , it is

experimentally seen that OC_2H_5^- ion, which is the conjugate base of $\text{C}_2\text{H}_5\text{OH}$ reacts fairly completely with H_2O to form $\text{C}_2\text{H}_5\text{OH}$ and OH^- ion.

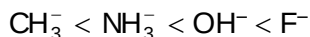


Ethoxide ion, $\text{C}_2\text{H}_5\text{O}^-$ is, therefore, a stronger base than OH^- and H_2O is a stronger acid than $\text{C}_2\text{H}_5\text{OH}$. Similarly when HS^- is added to NH_3 , it has been found experimentally that NH_4^+ and S^{2-} ions are present in the reaction mixture. This shows that NH_3 is a stronger base in comparison to HS^- .

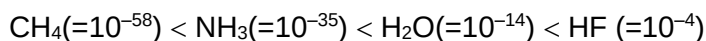
Periodic variations of acidic and basic properties

The discussion of this topic is made under the following heads.

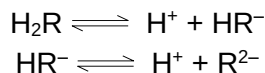
- a) **Hydracids of the elements of the same period.** We can consider the hydrides of the elements of 2nd period viz. CH_4 , NH_3 , H_2O and HF . These hydrides become increasingly acidic as we move from CH_4 to HF . Thus CH_4 has negligible acidic properties, but NH_3 donates a proton (H^+) to strong bases to form NH_2^- , H_2O loses a proton even more readily and HF is a fairly strongly acid. The increase in the acidic properties of these hydrides is due to the fact that as we move from CH_4 to HF , the stability of their conjugate base viz. CH_3^- , NH_2^- , OH^- and F^- increases in the order:



the increase in acidic properties is supported by the successive increase in the dissociation constant values of these hydrides as shown.



- b) **Hydracids of the elements of the same group.** The following examples make this point clear:
- Hydracids of VA groups elements (NH_3 , PH_3 , AsH_3 , SbH_3 , BiH_3). All these hydrides show basic character. With the increase in size and decrease in electronegativity from N to Bi, there is a decrease in electron density in sp^3 hybrid orbital and thus electron donor capacity (i.e. basic character) decreases.
 - Hydracids of VI A group elements (H_2O , H_2S , H_2Se , H_2Te). Aqueous solution of the hydrides of this group behave as weak diprotic acid and ionise as:

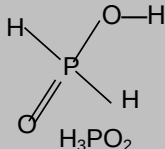
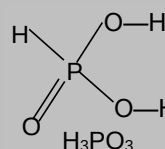
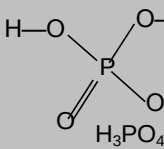


The strength of the hydrides as acids increases in the order:

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When the oxidation state rule as given above is applied to the oxyacids of phosphorus viz. $\text{H}_3\text{P}^+\text{O}_2 < \text{H}_3\text{P}^{+3}\text{O}_3 < \text{H}_3\text{P}^{+5}\text{O}_4$, but the experimental observation suggest the reverse order viz $\text{H}_3\text{PO}_2 \geq \text{H}_3\text{PO}_3 > \text{H}_3\text{PO}_4$.

Explanation: The experimental order can be explained when we consider the structures of these acids as given below. In these the number of protonated and unprotonated oxygen atoms have also been indicated. The oxygen atom attached with a proton is called protonated oxygen while that attached directly with phosphorus (central atom) is known as unprotonated oxygen.

Structure:			
Nature of the acid:	Monobasic	Dibasic	Tribasic
No. of protonated oxygen atoms	1	2	3
No. of unprotonated oxygen atoms:	1	1	1

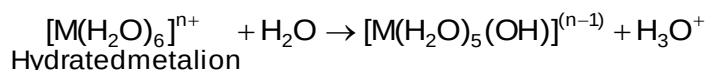
The proton attached to any oxygen atom has a far greater chance of dissociation than that linked directly with phosphorus atoms (which is the central atom). Thus in this series, since the number of protonated oxygen atoms and consequently the number of dissociable protons increases from one in H_3PO_2 to three in H_3PO_4 , the acidity of these acids decrease in the order : $\text{H}_3\text{PO}_2 \geq \text{H}_3\text{PO}_3 > \text{H}_3\text{PO}_4$

- ii) The acidic character of the oxyacids of different elements which are in the oxidation state decrease with the increase in atomic number of the central atom. The following series follow this rule

- $\text{HOCl}^+ > \text{HOBr}^+ > \text{HOI}^+$
- $\text{HCl}^{7+}\text{O}_4 > \text{HI}^{7+}\text{O}_4$
- $\text{H}_2\text{S}^{4+}\text{O}_3 > \text{H}_2\text{Se}^{4+}\text{O}_3$

Explanation: As the atomic number of the central atom increase, its electronegativity decreases and its size increase. As a result of this the tendency of the acid to lose a proton to water decrease. This makes the acid a weaker acid.

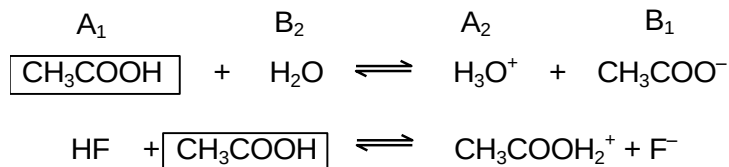
- d) **Hydrated metal ions:** Under favourable conditions one or more protons may dissociate from the coordinated aquo groups.



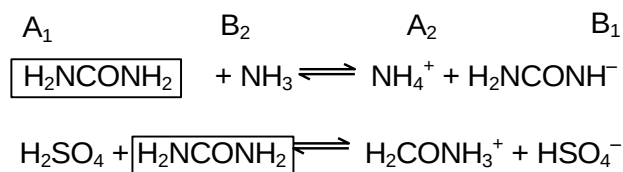
Thus hydrated metal ions also develop acidity. The other things being equal, acidity increases with the increase of positive charge and basicity increases with the increase of negative charge. Thus $[\text{Fe}(\text{H}_2\text{O})_6]^{3+}$ ion is a stronger acid than $[\text{Fe}(\text{H}_2\text{O})_6]^{2+}$ ion and $[\text{Ni}(\text{OH})_4]^{2-}$ is a stronger base than $[\text{Ni}(\text{OH})_4]^-$ ion.

Amphiprotic substances

These substances act as an acid as well as a base: e.g, CH_3COOH is acid with water while is a base with HF.

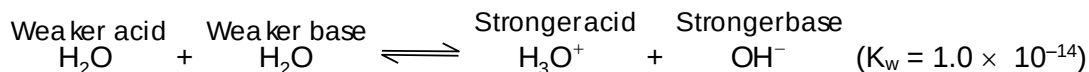


Urea is also amphiprotic as it is an acid with NH_3 while a base with sulphuric acid



Similarly water can act as an acid in the presence of bases stronger than itself such as NH_3 , amine, $\text{C}_2\text{H}_5\text{O}^-$, OH^- and CO_3^{2-} ions. Water can act as a base in the presence of acids stronger than itself such as HClO_4 , HCl , CH_3COOH and phenol.

In fact the amphiprotic nature of H_2O is well illustrated in the extremely slight dissociation or self-ionisation:

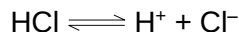
**The levelling effect, levelling and differentiating solvents**

The apparent strength of a protonic acid is dependent on the solvent in which the acid is dissolved. When all the acids in the acid chart which are stronger than H_3O^+ ion (i.e., the acids above H_3O^+ acids) are added to H_2O , they donate as proton to H_2O to H_3O^+ ion and appear to have equal strength, since all these acids are levelled to the strength of H_3O^+ ion which is left in solution and is common to all such solutions. This phenomenon viz. The strength of all the acids becomes equal to that of H_3O^+ ion is called leveling effect of the solvent, water, and here water is called a leveling solvent for all these acids.

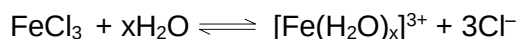
In aqueous solution all very strong bases like Na^+H^- , Na^+NH_2^- , $\text{Na}^+\text{OC}_2\text{H}_5^-$ are levelled to the strength of OH^- ion, for they react completely with H_2O to produce OH^- ions. The solvent in which complete proton-transfer occurs are called levelling solvents. In other words, the solvent in which the solute is ~100% ionised, are called levelling solvents. Since HF and HCl both are ~ 100% ionised in liquid NH_3 to give ~100% NH_4^+ ions, these appear to be of equal strength and liq. NH_3 acts as a levelling solvent for HF and HCl. In H_2O , HF is only partially ionised, whereas HCl and HBr are ~ 100% ionised. Thus H_2O is a differentiating solvent for HF, but for HCl and HBr it is a leveling solvent. Several mineral acids are partially ionised in glacial CH_3COOH medium because CH_3COOH is a poor proton-acceptor but rather a better proton donor. CH_3COOH , therefore, acts as a differentiating solvent towards the mineral acids. But, for bases, CH_3COOH act as a leveling solvent.

Utility of Bronsted Concept

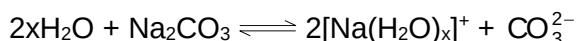
- i) It defines acids and bases in terms of the substances themselves and not in terms of their ions in aqueous solution. Thus HCl is an acid because of the fact that it can give a H^+ ion.



- ii) The Bronsted concept recognises that acid-base behaviour is neither restricted to, nor depends on, any particular, solvent
- iii) This concept is useful in accounting for the hydrolysis of salt solution. When a salt is dissolved in water, an unbalance in the concentration of the solvent cation (H_3O^+) and anion (OH^-) will result, if the salt cation and anion differ in their proton-donor and proton-acceptor properties towards H_2O . This point can be illustrated by considering the aqueous solution of $FeCl_3$. Aqueous hydrated ferric ion, $[Fe(H_2O)_x]^{3+}$, exceeds the proton-acceptor ability of Cl^- ion and a considerable excess of H_3O^+ ion is produced in the solution, making $FeCl_3$ acidic.

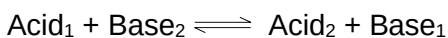


Aqueous solution of a Na_2CO_3 is alkaline in character, because the proton acceptor ability of CO_3^{2-} ion exceeds the proton-donor ability of hydrated sodium ion, $[Na(H_2O)_x]^+$.



Limitations

- i) This concept lays excessive emphasis on the proton - transfer. Although it is true that most common acids are protonic in nature, yet there are many which are not. It fails to explain the acidic character of many substances which do not contain any proton (e.g. SO_3 , $AlCl_3$).
- ii) There are a number of acid-base reactions in which no proton transfer takes place, e.g.
- $$SO_2 + SO_2 \rightleftharpoons SO^{2+} + SO_3^{2-}$$



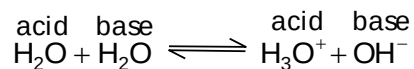
Thus the protonic definition cannot be used to explain the reactions occurring in non-protonic solvents such as $COCl_2$, SO_2 , N_2O_4 and BrF_3 .

3.3 General theory of solvent system

The protonic definition of acids and bases given by Bronsted can be extended to the reactions occurring in non-aqueous solvents containing hydrogen such as NH_3 , N_2H_4 , HF , H_2SO_4 , CH_3COOH , HCN and alcohols.

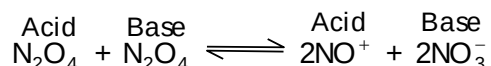
In an attempt to have a more general definition of acids and bases applicable to protonic and non-protonic solvents, several definitions have been proposed. One of these is due to Cady and Elsey (1928) according to whom an acid is solute that, either by direct dissociation or by reaction with the solvent gives the anion characteristic of the solvent and a base is a

solute that, either by direct association or by reaction with the solvent, gives the cation characteristic of the solvent. If for example, we consider the solvent H_2O , its characteristic cation and anion are H_3O^+ and OH^- respectively as shown below:



Thus all those compounds which can give H_3O^+ ions in H_2O will act as acids and all the compounds which can give OH^- ions in H_2O will behave as bases.

Similarly in N_2O_4 as solvent substance such as NOCl which yield NO^+ ions are acids and the substances such as NaNO_3 which yield NO_3^- ions are bases.



Evidently this definition of acids applies equally well to protonic and non-protonic solvents.

The auto-ionisation of some protonic and non-protonic solvents is shown below.

	Acid		Base		Acid		Base
Protonic solvents	H_2O	+	H_2O	$\rightleftharpoons$	H_3O^+	+	OH^-
	NH_3	+	NH_3	$\rightleftharpoons$	NH_4^+	+	NH_2^-
	CH_3COOH	+	CH_3COOH	$\rightleftharpoons$	$\text{CH}_3\text{COOH}_2^+$	+	CH_3COO^-
Non-protonic solvents	SO_2	+	SO_2	$\rightleftharpoons$	SO^{2+}	+	SO_3^{2-}
	COCl_2	+	COCl_2	$\rightleftharpoons$	$(\text{COCl}.\text{COCl}_2)^+$	+	Cl^-
	BrF_3	+	BrF_3	$\rightleftharpoons$	BrF_2^+	+	BrF_4^-
	N_2O_4	+	N_2O_4	$\rightleftharpoons$	2NO^+	+	2NO_3^-

Just as with the Arrhenius definition, neutralisation is a reaction between an acid and a base to produce a salt and the solvent. Neutralisation reaction in some non-aqueous solvents are given below:

	Acid			Base		Salt		Solvent
In liq. NH ₃	:	NH ₄ Cl	+	NaNH ₂	⇌	NaCl	+	2NH ₃
In N ₂ O ₄	:	NOCl	+	NaNO ₃	⇌	NaCl	+	N ₂ O ₄
In CH ₃ COOH	:	HCl	+	CH ₃ COONa	⇌	NaCl	+	CH ₃ OOH
In liq. SO ₂	:	SOCl ₂	+	[N(CH ₃) ₄] ₂ SO ₃	⇌	2[N(CH ₃) ₄]Cl	+	2SO ₂

It may be seen from the following reactions that there is a complete analogy between the solvolytic acid and amphoteric behaviours in aqueous solvents.

Solvolytic behaviour	In liq. NH_3 :	$\text{AlCl}_3 + \text{NH}_3 \longrightarrow [\text{Al}(\text{NH}_2)]^{2+} + \text{H}^+ + 3\text{Cl}^-$
	In H_2O :	$\text{AlCl}_3 + \text{H}_2\text{O} \longrightarrow [\text{Al}(\text{OH})]^{2+} + \text{H}^+ + 3\text{Cl}^-$
Amphoteric behaviour	In liq. NH_3 :	$\text{Zn}(\text{NH}_2)_2 + 2\text{NH}_2^- \longrightarrow [\text{Zn}(\text{NH}_2)_4]^{2-}$
	In H_2O	$\text{Zn}(\text{OH})_2 + 2\text{OH}^- \longrightarrow [\text{Zn}(\text{OH})_4]^{2-}$

Utility of the concept

Evidently this concept of solvent system can be used to explain the acid-base reactions occurring in aqueous solvents (protonic and non-protonic both). It also includes many non-aqueous acid-base systems.

Limitations

- This theory does not consider a number of acid base reactions included in the protonic definition.
- It limits acid base phenomena to solvent systems only. Thus it does not explain the acid - base reactions which may occur in the absence of solvent.
- It cannot explain the neutralisation reactions occurring without the presence of ions.

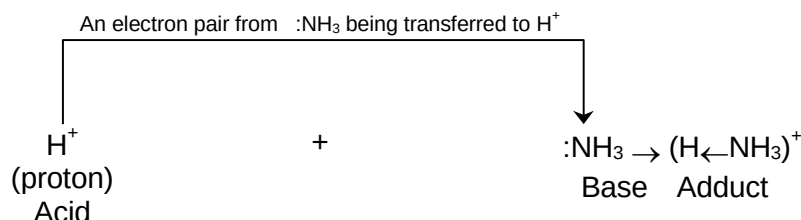
Thus this theory can simply be said to be an extension of the Arrhenius water - ion system.

3.4 The Lewis Concept - The Electron - Donor - Acceptor System

This theory explains the acid-base phenomena not in terms of ionic reactions but in terms of electronic structure of the acid and base along with the formation of a coordinate covalent bond. According to Lewis (1923), an acid is any species (molecule, radical or ion) that can accept an electron-pair to form a coordinate covalent bond and a base is any species that can donate an electron-pair to the formation of a coordinate covalent bond. Thus, in the Lewis system, an acid is an electron pair-acceptor and a base is an electron pair-donor.

Thus according to Lewis theory, the process of neutralisation is simply the formation of a coordinate bond between an acid and a base. The neutralisation product, termed as coordinate complex or adduct, may be either non-ionisable or may undergo dissociation or condensation reaction depending on its stability.

Now consider the reaction between a proton (H^+) and $:\text{NH}_3$ molecules as shown below.



Evidently in the above reaction proton (H^+) accepts one electron pair from $:\text{NH}_3$ molecule and is, therefore, acid, whereas $:\text{NH}_3$ molecule which donates an electron pair, is a base. The adduct is NH_4^+ ion.

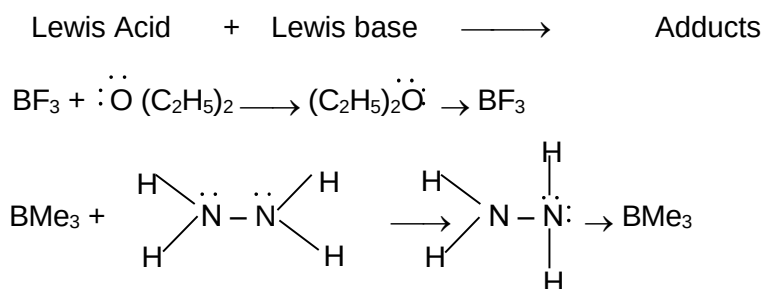
Lewis bases and Bronsted-Lowry bases are the same substances, since any molecule or ion which accepts protons does so because of the presence of an unshared pair of electrons. In the above example NH_3 molecule is a proton acceptor (i.e., Bronsted-base) and an electron pair donor (i.e., Lewis - base).

Bronsted and Lewis theories are thus identical as far as bases are concerned except that the wording used for definition of the bases is different in both the theories. Thus NH_3 , H_2O , OH^- , Cl^- , CN^- etc. are the bases on the Bronsted as well as on the Lewis systems. There are however, few compounds such as amides, ethers, nitriles, C_2H_4 , C_2H_2 , C_6H_6 etc. which have little or no tendency to accept protons but react readily with Lewis - acids.

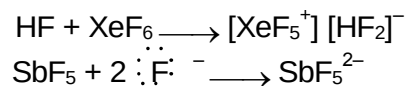
Classification of Lewis Acids

Any Lewis acid must contain at least one empty orbital in the valence shell of one of its atoms to accept an electron pair from a Lewis-base. Lewis - acids may be classified as:

- i) **Molecules containing a central atom with an incomplete octet.** Typical examples of this class of acids are electron deficient molecules such as alkyls and halides of Be, B and Al. Some reactions of this type of Lewis acid with Lewis bases are shown below:



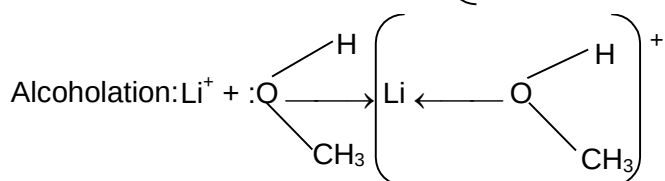
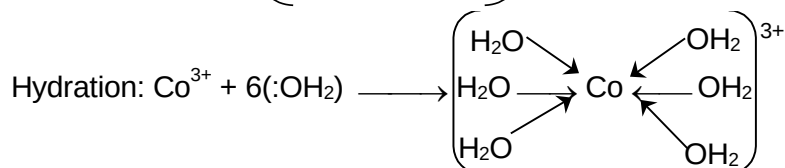
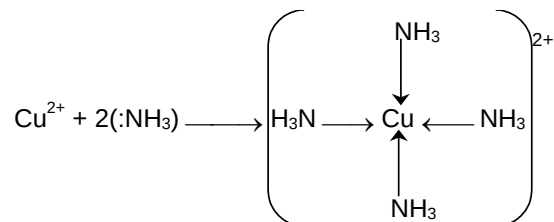
- ii) **Molecules containing a central atom with vacant d-orbitals.** The central atom of the halides such as SiX_4 , GeX_4 , TiCl_4 , SnX_4 , PX_3 , PF_3 , SF_4 , SeF_4 , TeCl_4 etc. have vacant d-orbitals. These substances can, therefore, accept an electron pair from the Lewis base to accommodate in their vacant d-orbital and can thus form adducts with a number of halide ions and organic bases. These substances are, therefore, Lewis acids. These halides are vigorously hydrolysed by H_2O to form an oxy acid or oxide of the central atom and the appropriate HX . The hydrolytic reactions take place presumably through the intermediate formation of unstable adducts with H_2O . For example



- iii) **Simple cations.** Theoretically all simple cations are potential Lewis acids. Reactions of some cations as Lewis acids with Lewis Bases are shown below. It will be seen that these reactions are identical with those which produce Werner complexes.

Lewis acid + Lewis base $\longrightarrow$ Adduct or Addition compounds

Ammonation: $\text{Ag}^+ + 2(\text{:NH}_3) \longrightarrow [\text{NH}_3 \longrightarrow \text{Ag} \longleftarrow \text{NH}_3]^+$



The Lewis acid strength or coordinating ability of the simple cations which, according to Lewis, are Lewis acids, increases with (a) an increase in the positive charge carried by the cation (b) an increase in the nuclear charge for atoms in any period of the periodic table. (c) a decrease in ionic radius. (d) a decrease in the number of shielding electron shells.

Evidently the acid strength of simple cations increases for the element on moving from left to right in a period and from bottom to top in a group of periodic table. Thus:

$\text{Fe}^{2+} < \text{Fe}^{3+}$ (positive charge increases, $+2 \longrightarrow +3$)

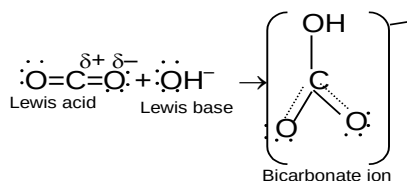
$\text{K}^+ < \text{Na}^+$ (on moving from bottom to top in a group)

$\text{Li}^+ < \text{Be}^{2+}$ (on moving from left to right in a period)

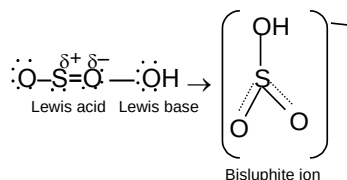
————— strength of Lewis acids increasing ————— $\longrightarrow$

iv) **Molecules having multiple bond between atoms of dissimilar electro-negativity.**

Typical examples of molecules falling in this class of Lewis acids are CO_2 , SO_2 and SO_3 . In these compounds the oxygen atoms are more electronegative than S- or C- atom. As a result, the electron density of π -electrons is displaced away from carbon or sulphur atoms which are less electronegative than oxygen, towards the O-atom. The C- or S-atom thus becomes electron deficient and is, therefore, able to accept an electron pair from a Lewis base such as OH^- ions to form dative bond.



SO₂ also reacts in the same manner with OH⁻ ion



- v) *Elements with an electron sextet.* Oxygen and sulphur atoms contain six electrons in their valence shell and can, therefore, be regarded as Lewis acids. The oxidation of SO₃²⁻ to SO₄²⁻ ion by oxygen and to S₂O₃²⁻ ion by sulphur are the acid-base reactions.

Lewis base		Lewis acid		Adduct
$\begin{array}{c} \text{O}^- \\ \\ :\text{S} = \text{O} \\ \\ \text{O}^- \end{array}$	+	$\ddot{\text{O}}:$	→	$\ddot{\text{O}} \leftarrow \begin{array}{c} \text{O}^- \\ \\ :\text{S} = \text{O} \\ \\ \text{O}^- \end{array}$ or SO ₄ ²⁻
$\begin{array}{c} \text{O}^- \\ \\ :\text{S} = \text{O} \\ \\ \text{O}^- \end{array}$	+	S	→	$\ddot{\text{S}} \leftarrow \begin{array}{c} \text{O}^- \\ \\ :\text{S} = \text{O} \\ \\ \text{O}^- \end{array}$ or S ₂ O ₃ ²⁻

Utility of Lewis concept

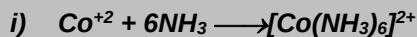
- This concept also includes those reactions in which no protons are involved.
- Lewis concept is more general than the Bronsted - Lowry concept (i.e. protonic concept) in that acid-base behaviour is not dependent on the presence of one particular element or on the presence or absence of a solvent.
- It explains the long accepted basic properties of metallic oxides and acidic properties of non-metallic oxides
- This theory also includes many reactions such as gas phase, high temperature and non-solvent reaction as neutralisation process.
- The Lewis approach is, however, of great value in case where the protonic concept is inapplicable, for example, in reaction between acidic and basic oxides in the fused state.

Limitations

- Since the strength of Lewis acids and bases is found to depend on the type of reaction, it is not possible to arrange them in any order of their relative strength. Thus, for example, experiments show that fluoride complex of Be²⁺ ions is more stable than that of Cu²⁺ ion, indicating that Be²⁺ ion is more acidic than Cu²⁺ ion. On the other hand amine complex of Cu²⁺ is more stable than that of Be²⁺ ion indicating that Cu²⁺ is more acidic than Be²⁺ ion.

- ii) According to the phenomenological criteria, an acid-base reaction should be a rapid reaction. There are, however, many Lewis acid-base reactions which are slow.

Exercise 2: In the following reaction, identify each of the reactant as a Lewis acid or Lewis base.



3.5 Hard and Soft Acids and Bases

The Lewis definition recognizes acids and bases in terms of their ability to accept or donate electron pairs. As such, the strength of any acid or base is determined by the very nature of the reaction involved in any particular electron transfer process. Accordingly, assignment of any single consistent criterion for acid base strength becomes very difficult in the Lewis definition. Attempts have been made to correlate various strength factors from enthalpy changes (ΔH) for acids – base combinations.

However, a qualitative correlation between the various Lewis acids and bases has been achieved. This is done by classifying the acids or bases into two classes hard and soft. Acid – base reactions are then treated by the general principle that hard acids prefer to combine with hard bases and soft acids prefer to combine with soft bases (and vice versa).

The criterion of hardness (or softness) is ascribed to the “hardness” of the electron cloud associated with any species. A firmly held electron–cloud with low polarisability makes a species “hard”, while an easily polarisable electron cloud characterises the species as “soft”. A third category with intermediate character appears in the border line. The principal distinguishing features of hard and soft acid and bases may then be summarised as follows:

Hard acids	Soft acids
i) Small size	i) large size
ii) high positive oxidation state absence of any outer electron which are easily excited to higher states	ii) zero or low positive oxidation state several easily excitable valence electrons
$\text{H}^+, \text{Li}^+, \text{Na}^+, \text{K}^+$	$\text{Cu}^+, \text{Ag}^+, \text{Hg}^+$
$\text{Be}^{2+}, \text{Mg}^{2+}, \text{Ca}^{2+}, \text{Sr}^{2+}$	$\text{Cd}^{2+}, \text{Pt}^{2+}, \text{Hg}^{2+}, \text{Pt}^4$
$\text{Al}^{3+}, \text{Fe}^{3+}, \text{Co}^{3+}, \text{Cr}^{3+}$	$\text{I}^+, \text{I}_2, \text{Br}_2, \text{Br}^+$
CO_2, SO_3	$\text{Cl}, \text{Br}, \text{I}, \text{N}$
HX (hydrogen –bonding molecules)	M^0 (metal atoms) and bulk metals

Hard Bases	Soft bases
i) high electronegativity	i) low electronegativity
iii) Presence of filled orbitals; empty orbitals may exist at high energy level	ii) high polarisability (iii) partially filled orbitals; empty orbitals are low – lying.

$\text{H}_2\text{O}, \text{OH}^-, \text{F}^-, \text{Cl}^-$	H^-, I^-
$\text{CH}_3\text{CO}_2^-, \text{PO}_4^{3-}, \text{SO}_4^{2-}$	$\text{SCN}^-, \text{CN}^-, \text{S}_2\text{O}_3^{2-}$
$\text{NO}_3^-, \text{CO}_3^{2-}, \text{ClO}_4^-$	$\text{R}_2\text{S}, \text{RSH}, \text{RS}^-$
$\text{NH}_3, \text{N}_2\text{H}_4, \text{RNH}_2$	$\text{R}_3\text{P}, \text{R}_3\text{As}.$

Exercise 3: Identify hard and soft acids and bases among the following $\text{CO}_2, \text{CO}_3^{2-}, \text{I}^+, \text{I}^-, \text{CN}^-, \text{Fe}^{+3}, \text{Br}_2$

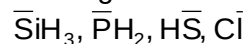
Illustration 1: Identify the most acidic oxide
 $\text{Al}_2\text{O}_3, \text{P}_4\text{O}_{10}, \text{SiO}_2, \text{Sb}_2\text{O}_5$

Solution: The oxide with the highest positive oxidation state on the atom other than oxygen should be most acidic. Oxidation state of Sb in Sb_2O_5 and P in P_4O_{10} + 5. But, due to higher electronegativity of the P than Sb, P_4O_{10} is the most acidic oxide.

Illustration 2: Predict the relative acidic strength among the following, $\text{SiH}_4, \text{PH}_3, \text{H}_2\text{S}, \text{HCl}$.

Solution: Assume that each of these have lost a proton

So we get



We will qualitatively analyse the charge density in each of the species and we will follow the rule given below.

- Larger the volume over which the charge is spread.
- Smaller is the charge density.
- Smaller is the basic character of the ion to attract a proton.
- Larger is the acidity of the conjugate acid.

In the above case one can see that, the size of the central atom over which the negative charge is present is decreasing from left to right. But 1/4 volume of silicon is overlapped by hydrogens in $\overline{\text{SiH}}_3$, 2/3 of the volume of P is overlapped by hydrogens in $\overline{\text{PH}}_2$, 1/2 the volume of sulphur is overlapped by hydrogens in $\overline{\text{HS}}$ whereas, the entire volume of Cl is available for the charge in $\overline{\text{Cl}}$. So, actually the space available for the charge is increasing in the order $\overline{\text{SiH}}_3 < \overline{\text{PH}}_2 < \overline{\text{HS}} < \overline{\text{Cl}}$. Therefore, the following conclusion may be arrived at

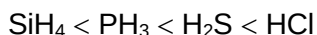


Illustration 4: Predict the relative acidic strength among the following :
 H_2O, H_2S, H_2Se, H_2Te .

Solution: Assume that each has lost a proton : So, we get : HO^- , HS^- , HSe^- , HTe^-
 It can be easily seen that the volume available for the negative charge is increasing from HO^- to HTe^- , Therefore :

1. Volume available for the negative charge is increasing from left to right.
2. Charge density is decreasing from left to right.
3. Basicity is decreasing from left to right.
4. Acidity of conjugate acids is increasing from left to right.

4. Origin of Acidity and Basicity in Organic Compounds

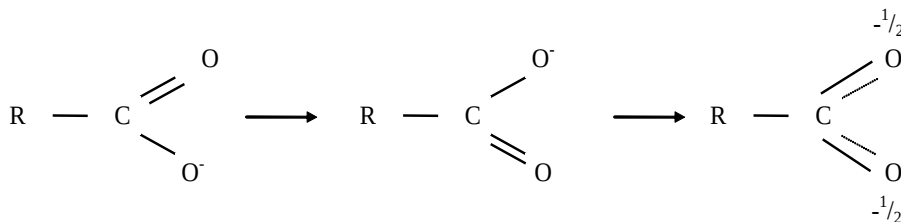
1. Among hydrocarbons, acidity increases with the % 's' character . This is because, higher the 's' character, closer are the electrons to the nucleus of C and farther the electrons go from H, easily can H be removed as H^+ .

Therefore $CH_3 - CH_3 < CH_2 = CH_2 < CH \equiv CH$

2. Alcohols are more acidic than Alkanes because oxygen is more electro negative than C and consequently O - H bond easily give H^+ than C - H bond

$H_3C - H < CH_3 - O - H$

3. Carboxylic acids are more acidic than alcohols because carboxylate anion is more resonance stabilised than carboxylic acid as compared to alkoxide anion with respect to alcohol.



Therefore $CH_3O - H < H - \overset{\overset{O}{\parallel}}{C} - OH$

4. Phenols are more acidic than alcohols because phenoxide ion is more resonance stabilised as compared to phenol than alkoxide as compared to alcohol.

Therefore $CH_3 - O - H < C_6H_5 - O - H$

5. Phenols are less acidic than carboxylic acids. This is because in phenoxide ion some of the resonance structures are of higher energy and this makes phenoxide ion not very much stable as compared to phenol.

Therefore $C_6H_5 O - H < H - \overset{\overset{O}{\parallel}}{C} - OH$

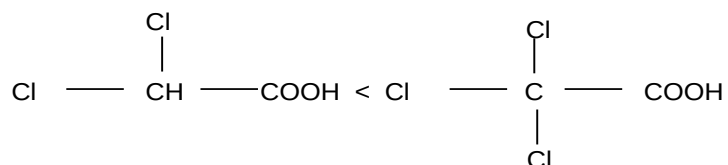
6. In simple aliphatic acids, more the no. of CH_3 added, less is the acidity. This is due to the +I (inductive) effect of CH_3 . This effect of adding CH_3 decreases with increase in distance between CH_3 added to the substituent, which releases H^+ .



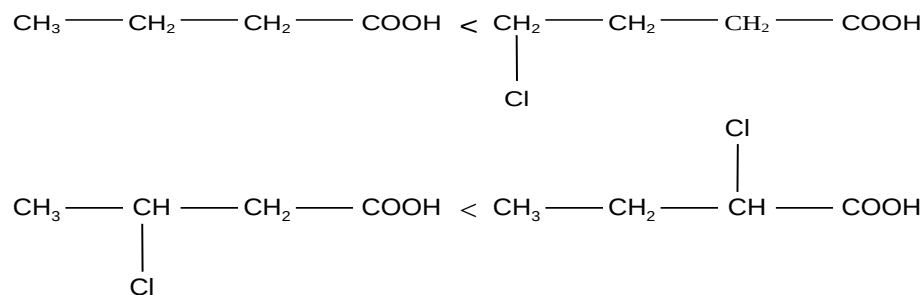
7. Increase in 's' character increases acidity of carboxylic acids.



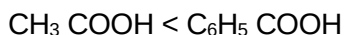
8. Increase in electron withdrawing substituents into simple aliphatic acids increases acidity.



9. Nearer Cl is placed to COOH group, more is acidity.



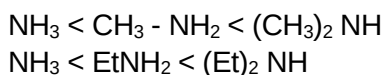
10. Benzoic acid is more acidic than carboxylic acids because benzoate anion shows/has more resonance.



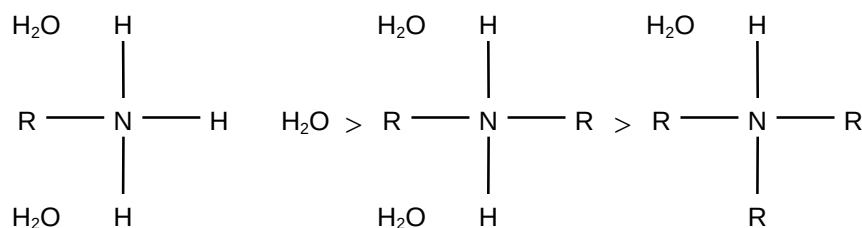
Exception to this is formic acid.

There is no destabilising factors for formate anion. Therefore, it is highly stabilised compared to formic acid because of resonance. Where as, Benzoic acid itself is resonance stabilised due to benzene ring. Therefore, benzoate ion is not very stable compared to benzoic acid. Thus $\text{C}_6\text{H}_5\text{COOH} < \text{HCOOH}$

11. More the no. of methyl or ethyl groups in NH_3 , more is its basicity. This is because of +I effect of methyl or ethyl group which helps nitrogen to donate its lone pair of electrons.



An exception to the above trend is the trisubstituted derivative. It is seen that if we introduce an alkyl group in a secondary amine, the basic strength of amines actually decreases. This is due to the fact that, the basic strength of an amine in water is determined not only by electron - availability on the nitrogen atom, but also by the extent to which the cation, formed by uptake of a proton, can undergo solvation, and so become stabilised. The more the number of hydrogen atoms attached to nitrogen in the cation, the greater the possibilities of powerful solvation via hydrogen bonding between these and water which increases stabilisation by solvation.



Thus, on going along the series $\text{NH}_3 \rightarrow \text{RNH}_2 \rightarrow \text{R}_2\text{NH} \rightarrow \text{R}_3\text{N}$, the inductive effect will tend to increase the basicity, but progressively less stabilisation of the cation by hydration will occur, which will tend to decrease the basicity. The net effect of introducing successive alkyl groups thus becomes progressively smaller, and an actual change over takes place, on going from secondary to a tertiary amine. That is why, $(\text{CH}_3)_3\text{N} < (\text{CH}_3)_2\text{NH}$

If this is the explanation, no such changeover should be observed, if measurements of basicity are made in solvent in which hydrogen- bonding cannot take place; it has, indeed, been found that, in chlorobenzene the order of basicity of the butylamines is



12. Aniline is less basic than NH_3 . This is because in aniline the nitrogen atom is again bonded to an sp^2 hybridised carbon atom but, more significantly, the unshared electron pair on nitrogen can interact with the delocalised orbital of the nucleus:

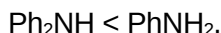
If aniline is protonated, any such interaction, with resultant stabilisation in the anilinium cation is prohibited, as the electron pair on N is no longer available :

The aniline molecule is thus stabilised with respect to the anilinium cation, and it is therefore energetically unprofitable for aniline to take up a proton ; it thus functions as a base with atmost reluctance.

13. Introduction of alkyl, e.g. Me groups on the nitrogen atom of aniline results in small increase in pK_a due to the +I effect of alkyl groups.



14. Introduction of phenyl groups on N lowers basicity because the substituted amine becomes much more stable than the ion.



5. Solution to Exercises

Exercise 1:	Conjugate Acid		Conjugate Base	
	CH_3NH_2		CH_3NH^-	
	H_2O		OH^-	
	CH_3COOH		CH_3COO^-	
	NH_4^+		NH_3	
Exercise 2:	Lewis acid	Co^{2+}	Lewis Base	NH_3
	Lewis acid	BF_3	Lewis Base	NH_3
Exercise 3:	Hard Acids	$\text{CO}_2, \text{Fe}^{3+}$	Soft Acid	I^+, Br_2
	Hard Bases	CO_3^{2-}	Soft bases	I^-, CN^-

6. Solved Problems

6.1 Subjective

Problem 1: Why sulphuric acid is stronger than sulphurous acid?

Solution: Subphuric acid is stronger than sulphurous since the sulphur atom in sulphuric acid has more co-ordinated oxygen atoms.



Problem 2: Explain
 F^- is a strong base

Solution: F^- is a conjugate base of acid HF. HF is a weak acid. Weaker the acid, stronger the conjugate base.

Problem 3: In hypophosphorus acid P is in +1 state (H_3PO_2) and in phosphoric acid (H_3PO_4) P is in +5 state but still H_3PO_2 is stronger than H_3PO_4 . Explain.

Solution: A consideration of the structures of these acids explains the observed trend.



The hydrogen atoms bonded directly to phosphorus are not acidic as the P – H bond is not polar enough due to equal electronegativity of phosphorus and hydrogen. Acidity arises due to O – H groups, one in H_3PO_2 and three in

H_3PO_4 . The inductive effect of the single oxygen atom acts in the single $\text{O}-\text{H}$ bond in H_3PO_2 but it is distributed over three $\text{O}-\text{H}$ bonds in H_3PO_4 respectively. So $\text{O}-\text{H}$ bond in H_3PO_2 much easily broken, explaining its higher acidity over H_3PO_2 .

Problem 4: Compare the acidity between HClO_4 and H_3AsO_3 .

Solution: HClO_4 is much stronger than H_3AsO_3 , conjugate base of it more stabilised than conjugate base of H_3AsO_3 .

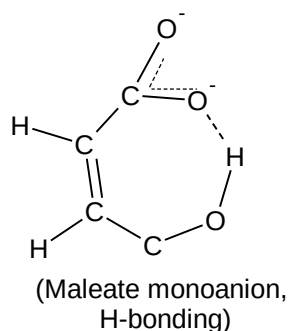
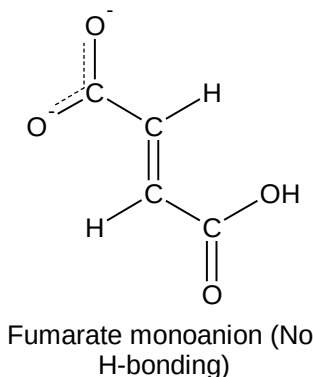
Problem 5: *p*-nitro phenol is stronger acid than phenol. Explain

Solution: Conjugate base of *p*-nitrophenol is more resonance stabilized than conjugate base of phenol (one more canonical form is possible for conjugate base of *p*-nitro phenol).

Problem 6: On the basis of H-bonding explain that the second ionization constant K_2 for fumaric acid is greater than for maleic acid.

Solution: We know that H-bonding involving acidic H has an acid weakening effect and H-bonding in conjugate base has an acid strengthening effect.

Both dicarboxylic acids have two ionisable H-atoms considering second ionization step.



Problem 7: Arrange *o*-nitro, *p*-nitro, *m*-nitro benzoic acids in increasing acid strength.

Solution: *o*-nitrobenzoic acid < *p*-nitrobenzoic acid < *m*-nitro benzoic acid
Nitro group ($-\text{NO}_2$) is electron withdrawing group and thus strengthen the acid. Its resonance effect, which occurs only from para and ortho positions, predominates over its inductive effect. Which occurs also from the meta position between ortho and para derivative.

Problem 8: Which is a stronger base towards a proton NH_2^- or PH_2^- and why?

Solution: Bond energy ($\text{N}-\text{H} > \text{P}-\text{H}$) suggests that NH_2^- will be stronger base. This is constituent with the relative strengths of the respective conjugate acids: $\text{NH}_3 < \text{PH}_3$.

Problem 9: Compare the basic strengths of ortho, meta and para isomers of nitroaniline and predict the decreasing order of basicity.

Solution: The $-\text{NO}_2$ group which is a powerful electron withdrawing inductive effect when present at ortho and para positions for the interaction of the unshared pair of the amino nitrogen with the delocalised p-orbital system of the benzene nucleus is then enhanced. The neutral molecule is thus stabilized even further with respect to the cation, resulting in further weakening as a base. Thus the order of basicity of nitroanilines is as



Problem 10: Which way would the pH shift when AlCl_3 is added to pure water? Write a net ionic equation to support your answer.

Solution: $\text{AlCl}_3 + 3\text{H}_2\text{O} \rightleftharpoons \text{Al}(\text{OH})_3 + 3\text{H}^+ + 3\text{Cl}^-$
As it is clear from the above equation that aqueous solution of AlCl_3 is acidic, so pH will shift below 7.

6.2 Objective

Problem 1: The following acids have been arranged in order of increasing acid strength. Identify the correct order.

ClOH (I) BrOH (II) IOH (III)

(A) $\text{I} < \text{II} < \text{III}$

(B) $\text{II} < \text{I} < \text{III}$

(C) $\text{III} < \text{II} < \text{I}$

(D) $\text{I} < \text{III} < \text{II}$

Solution: Higher the electronegativity of the atom attached with oxygen greater the acidity. So order is $\text{III} < \text{II} < \text{I}$
 $\therefore$ (C)

Problem 2: Among the following compounds, the strongest acid is

(A) $\text{HC} \equiv \text{CH}$

(B) C_6H_6

(C) C_2H_6

(D) CH_3OH

Solution: Hydrogen atom attached with oxygen is maximum acidic.
 $\therefore$ (D)

Problem 3: The conjugate acid of HPO_4^{2-} is

(A) PO_4^{3-}

(B) H_2PO_4^-

(C) H_3PO_4

(D) HPO_3^-

Solution: Conjugate acid and base differs by one proton hence the conjugate acid of HPO_4^{2-} is H_2PO_4^-
 $\therefore$ (B)

Problem 4: The anhydride of HClO_4 is

(A) ClO_2

(B) Cl_2O_3

(C) Cl_2O_7

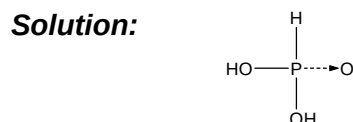
(D) None

Solution: Cl has +7 oxidation state in Cl_2O_7 and HClO_4
 $\therefore$ Anhydride is Cl_2O_7
 $\therefore$ (C)

Problem 5: The conjugate acid of azide ion is
 (A) NH_3 (B) N_3H
 (C) N_2H_4 (D) N_2^-

Solution: $\text{N}_3^- + \text{H}^+ \longrightarrow \text{N}_3\text{H}$
 $\therefore$ (B)

Problem 6: H_3PO_3 is acid
 (A) Monobasic (B) Dibasic
 (C) Tribasic (D) None

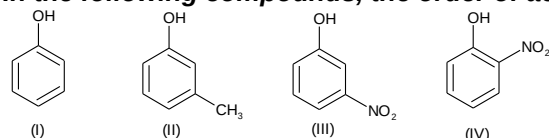


Thus two hydrogens are attached with oxygen, therefore H_3PO_3 is dibasic acid
 $\therefore$ (B)

Problem 7: Which salt undergoes hydrolysis
 (A) KCl (B) NaNO_3
 (C) CaSO_4 (D) HCOONH_4

Solution: Salts of strong acids and strong bases do not hydrolyse. Thus only HCOONH_4 is hydrolysed
 $\therefore$ (D)

Problem 8: In the following compounds, the order of acidity is



(A) $\text{III} > \text{IV} > \text{I} > \text{II}$ (B) $\text{I} > \text{IV} > \text{III} > \text{II}$
 (C) $\text{II} > \text{I} > \text{III} > \text{IV}$ (D) $\text{IV} > \text{III} > \text{I} > \text{II}$

Solution: +I effect decreases acidity, -I and -R effect increases acidity. NO_2 group cannot exert -R effect from the meta position.
 $\therefore$ (D)

Problem 9: Which of the following species is an acid and also a conjugate base of another acid
 (A) HSO_4^- (B) H_2SO_4
 (C) OH^- (D) H_3O^+

Solution: HSO_4^- can donate a proton also can accept a proton.
 $\therefore$ (A)

Problem 10: Among the following which is the least basic
 (A) F^- (B) I^-
 (C) Cl^- (D) Br^-

Solution: HI is the strongest acid among the hydro acids of halogen, so I^- is least acid.

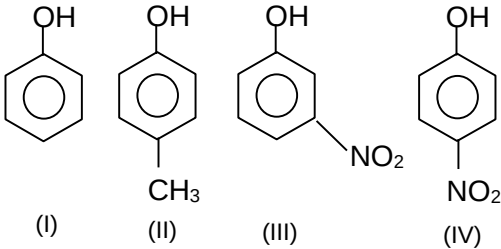
7. Assignments

7.1 Subjective

LEVEL - I

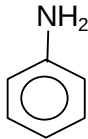
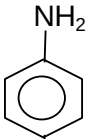
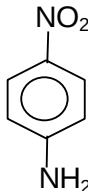
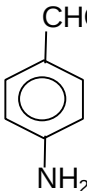
- Write down the conjugate base of the following
(i) NH_3 , (ii) CH_3COOH (iii) OH^- , (iv) $\text{H}_2\text{NCONH}_3^+$
- Predict the relative basic strength among the following
 HOI , HOIO , HOIO_2 , HOIO_3
- Arrange according to increasing Lewis acid character,
 SiF_4 , SiCl_4 , SiBr_4 , SiI_4
- The ionization constant of HCN is given as 4×10^{-10} . The ionization constant of NH_3 is 1.8×10^{-5} . Decide which is the stronger acid, HCN or NH_4^+ .
- Between Na^+ and Ag^+ which is a stronger Lewis acid and why
- Arrange the following in the order of their decreasing acidic strength
 CH_4 , LiH , BeH_2 , H_2O , HF , NH_3
- What should be the order of acidic strength in the series H_3PO_4 , H_3PO_3 and H_3PO_2 ?
- Which way would the pH shift when CuSO_4 is added to pure water. Write a net ionic equation to support your answer.
- Predict the relative acidic strength among the following:
 HIO_4 , HI , ICl
- Arrange according to increasing Lewis acid character,
 $\text{B}(\text{n-Bu})_3$, $\text{B}(\text{t-Bu})_3$

LEVEL - II

- Arrange the following in correct order of acidity
 CaO , SiO_2 , SO_3 , P_2O_5
 I II III IV
- What is the order of increasing strength of α -bromo butanoic acid, β -bromo butanoic acid, γ -bromo butanoic acid n – butanoic acid.
- Why pyrole is less basic than pyridine
- Arrange the following compounds in the increasing order of their basic strength.
 $\text{CH}_3\text{NH}^-\text{Na}^+$, $\text{C}_2\text{H}_5\text{NH}_2$, $(\text{iso-C}_3\text{H}_7)_3\text{N}$ and CH_3CONH_2
- Write the cation, anion and the neutralisation reaction if thionyl chloride reacts with sodium sulphate.
- Arrange the following in the order of their basic strength. Justify your answer
 NH_2NH_2 , NH_3 , NH_2OH
- Arrange the following in correct order of acidity

 (I) (II) (III) (IV)
- Why diphenyl amine is less basic than aniline
- Arrange the following in decreasing order of basicity
 Phenol, o-nitrophenol, cresol
- Phenol is acidic but why ethanol is neutral

7.2. Objective

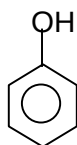
LEVEL – I

- Which one of following is the strongest acid
 (A) ClO_3^- (B) ClO_2^-
 (C) ClO_4^- (D) ClO^-
- The conjugate acid of HS_2O_7^- is
 (A) $\text{H}_2\text{S}_2\text{O}_7$ (B) H_2SO_4
 (C) HS_2O_6^+ (D) HSO_4^-
- Among the following weakest acid is
 (A) HNO_3 (B) H_3AsO_4
 (C) H_3SbO_4 (D) H_3PO_4
- Increasing order of Lewis acid strength
 AlCl_3 GaCl_3 InCl_3
 I II III
 (A) $\text{I} < \text{II} < \text{III}$ (B) $\text{III} < \text{II} < \text{I}$
 (C) $\text{II} < \text{I} < \text{III}$ (D) $\text{III} < \text{I} < \text{II}$
- Which one of following is the strongest acid
 (A) H_3PO_4 (B) H_3PO_2
 (C) H_3PO_3 (D) H_2SO_3
- What is the decreasing order of strength of bases
 (A) $\text{CH}_3 - \text{CH}_2^- > \text{NH}_2^- > \text{H} - \text{C} \equiv \text{C}^- > \text{OH}^-$
 (B) $\text{H} - \text{C} \equiv \text{C}^- > \text{CH}_3 - \text{CH}_2^- > \text{NH}_2^- > \text{OH}^-$
 (C) $\text{OH}^- > \text{NH}_2^- > \text{H} - \text{C} \equiv \text{C}^- > \text{CH}_3 - \text{CH}_2^-$
 (D) $\text{NH}_2^- > \text{H} - \text{C} \equiv \text{C}^- > \text{OH}^- > \text{CH}_3 - \text{CH}_2^-$
- Which of the following is a soft base?
 (A) R_2S (B) NH_3
 (C) H_2O (D) Cu^-
- Amongst the following the most basic compound is
 (A)  (B) 
 (C)  (D) 

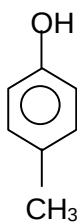
9. Which of the following is strongest Lewis acid
(A) CH_3^- (B) F^-
(C) NH_2^- (D) OH^-
10. The conjugate base of H_3O^+ is
(A) H_2O (B) OH^-
(C) H^+ (D) None of these
11. The conjugate acid of $\text{S}_2\text{O}_8^{2-}$
(A) $\text{H}_2\text{S}_2\text{O}_8$ (B) H_2SO_4
(C) HSO_4^- (D) HS_2O_8^-
12. The following acids have been arranged in order of decreasing acid strength. Identify the correct order.
 ClOH (I) BrOH (II) IOH (III)
(A) $\text{I} > \text{II} > \text{III}$ (B) $\text{II} > \text{I} > \text{III}$
(C) $\text{III} > \text{II} > \text{I}$ (D) $\text{I} > \text{III} > \text{II}$
13. Which of the following are amphiprotic in nature?
(A) OH^- (B) H_2PO_4^-
(C) HSO_3^- (D) HF
14. Among the following compounds, the strongest acid is
(A) $\text{HC} \equiv \text{CH}$ (B) C_6H_6
(C) C_2H_6 (D) CH_3OH
15. With reference to protonic acid which of the following statement is correct ?
(A) PH_3 is more basic than NH_3 (B) PH_3 is less basic than NH_3
(C) PH_3 is equally basic than NH_3 (D) PH_3 is amphoteric while NH_3 is basic
16. Acetic acid is a conjugate base of which acid.
(A) CH_3COO^- (B) $\text{CH}_3\text{CH}_2\text{COO}^-$
(C) $\text{CH}_3\text{COOH}_2^+$ (D) None
17. Conjugate base of NH_3 is
(A) NH_2^- (B) NH_4^+
(C) NH_4OH (D) NH_2OH
18. Which of the following is the soft base?
(A) CO_3^{2-} ion (B) CO
(C) Ce^{3+} (D) Pb^{2+}
19. Which among the following is the strongest base
(A) $\text{Be}(\text{OH})_2$ (B) $\text{Mg}(\text{OH})_2$
(C) $\text{Ba}(\text{OH})_2$ (D) $\text{Sr}(\text{OH})_2$
20. The conjugate acid of NH_3 is
(A) NH_2^- (B) NH_2OH



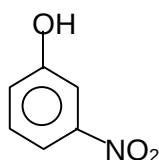
21. In the equilibrium $\text{CH}_3\text{COOH} + \text{HF} \rightleftharpoons \text{CH}_3\text{COOH}_2^+ + \text{F}^-$
 (A) F^- is the conjugate acid of CH_3COOH
 (B) F^- is the conjugate base of HF
 (C) CH_3COOH is the conjugate acid of $\text{CH}_3\text{COOH}_2^+$
 (D) $\text{CH}_3\text{COOH}_2^+$ is the conjugate base of CH_3COOH .
22. Which of the following statements is/are correct?
 (A) Maleic acid is a stronger acid than fumaric acid but maleate monoanion is a weaker acid than fumarate monoanion
 (B) Maleic acid is a stronger acid than fumaric acid and also maleate monoanion is a stronger acid than fumarate monoanion
 (C) Maleic acid is a weaker acid than fumaric acid but maleate monoanion is a stronger acid than fumarate monoanion
 (D) Maleic acid is a weaker acid than Fumaric acid and also maleate monoanion is a weaker acid than fumarate monoanion.
23. Which of the following is the strongest acid
 (A) HNO_3 (B) H_2SO_4
 (C) HNO_2 (D) All have same strength
24. Which of the following species is an acid and also a conjugate base of another acid
 (A) HSO_4^- (B) H_2SO_4
 (C) OH^- (D) H_3O^+
25. Which of the following orders regarding acid strength is correct?
 (A) $\text{HCOOH} < \text{CH}_3\text{COOH} < \text{PhCOOH}$ (B) $\text{HCOOH} > \text{PhCOOH} > \text{CH}_3\text{COOH}$
 (C) $\text{HCOOH} > \text{CH}_3\text{COOH} > \text{PhCOOH}$ (D) $\text{CH}_3\text{COOH} > \text{HCOOH} > \text{PhCOOH}$
26. The following equilibrium is established when hydrogen chloride is dissolved in acetic acid.
 $\text{HCl} + \text{CH}_3\text{COOH} \longrightarrow \text{Cl}^- + \text{CH}_3\text{COOH}_2^+$
 The set that characteristics the conjugate acid-base pair is
 (A) $(\text{HCl}, \text{CH}_3\text{COOH})$ and $(\text{CH}_3\text{COOH}_2^+, \text{Cl}^-)$
 (B) $(\text{HCl}, \text{CH}_3\text{COOH}_2^+)$ and $(\text{CH}_3\text{COOH}, \text{Cl}^-)$
 (C) $(\text{CH}_3\text{COOH}_2^+, \text{HCl})$ and $(\text{Cl}^-, \text{CH}_3\text{COOH})$
 (D) $(\text{HCl}, \text{Cl}^-)$ and $(\text{CH}_3\text{COOH}_2^+, \text{CH}_3\text{COOH})$
27. In the following compounds, the order of acidity is



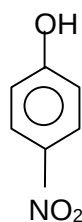
(I)



(II)



(III)



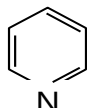
(IV)

- (A) III > IV > I > II
(B) I > IV > III > II
(C) II > I > III > IV
(D) IV > III > I > II
28. Of the following orders regarding basicity, which one is correct
(A) $\text{CH}_3\text{CH}_2\text{NH}_2 > \text{PhCONH}_2 > \text{CH}_3\text{CONH}_2$
(B) $\text{CH}_3\text{CH}_2\text{NH}_2 > \text{PhCONH}_2 > \text{CH}_3\text{CONH}_2$
(C) $\text{CH}_3\text{CH}_2\text{NH}_2 > \text{PhCONH}_2 < \text{CH}_3\text{CONH}_2$
(D) $\text{CH}_3\text{CH}_2\text{NH}_2 < \text{PhCONH}_2 < \text{CH}_3\text{CONH}_2$

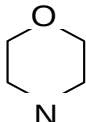
29. In the following compounds, the order of basicity



(I)



(II)



(III)



(IV)

- (A) IV > I > III > II
(B) III > I > IV > II
(C) II > I > III > IV
(D) I > III > II > IV

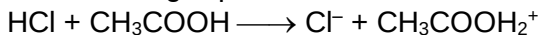
30. The conjugate base of $[\text{Cr}(\text{H}_2\text{O})_5\text{Cl}]^{2+}$ is
(A) $[\text{Cr}(\text{H}_2\text{O})_5]^{3+}$
(B) $[\text{Cr}(\text{H}_2\text{O})_4\text{Cl}]^{2+}$
(C) $[\text{Cr}(\text{H}_2\text{O})_4\text{OHCl}]^{3+}$
(D) $[\text{Cr}(\text{H}_2\text{O})_4\text{OHCl}]^+$

LEVEL - II**More than one choice**

1. Which of the following is a hard acid
(A) CO_2
(B) Br_2
(C) Fe^{3+}
(D) Cd^{2+}
2. Which of the following statements is/are correct
(A) I^- is a weaker base than F^-
(B) $\text{H}\ddot{\text{O}}^-$ is a stronger base $\text{H}_2\ddot{\text{N}}^-$
(C) HONH_2 is a weaker base than NH_3
(D) F_3C^- is a stronger base than Cl_3C^-
3. In the equilibrium $\text{CH}_3\text{COOH} + \text{HF} \rightleftharpoons \text{CH}_3\text{COOH}_2^+ + \text{F}^-$, which of the following statement(s) is/are correct:
(A) F^- is the conjugate acid of CH_3COOH
(B) F^- is the conjugate base of HF
(C) CH_3COOH is the conjugate acid of $\text{CH}_3\text{COOH}_2^+$
(D) $\text{CH}_3\text{COOH}_2^+$ is the conjugate base of CH_3COOH .
4. Which of the following statements is/are not correct?
(A) Maleic acid is a stronger acid than fumaric acid but maleate monoanion is a weaker acid than fumarate monoanion
(B) Maleic acid is a stronger acid than fumaric acid and also maleate monoanion is a stronger acid than fumarate monoanion

- (C) Maleic acid is a weaker acid than fumaric acid but maleate monoanion is a stronger acid than fumarate monoanion
 (D) Maleic acid is a weaker acid than Fumaric acid and also maleate monoanion is a weaker acid than fumarate monoanion.

5. The following equilibrium is established when hydrogen chloride is dissolved in acetic acid.



The set that characterises the conjugate acid-base pair is/are not correct:

- (A) (HCl, CH₃COOH) and (CH₃COOH₂⁺ and Cl⁻)
 (B) (HCl, CH₃COOH₂⁺) and (CH₃COOH, Cl⁻)
 (C) (CH₃COOH₂⁺, HCl) and (Cl⁻, CH₃COOH)
 (D) (HCl, Cl⁻) and (CH₃COOH₂⁺, CH₃COOH)

LEVEL - III

Other Engg. Exams

1. The following equilibrium is established when HCl is dissolved in acetic acid:



The set that characterizes the conjugate acid-base pair is

- (A) (HCl, CH₃COOH) and (CH₃COOH₂⁺, Cl⁻)
 (B) (HCl, CH₃COOH₂⁺) and (CH₃COOH, Cl⁻)
 (C) (CH₃COOH₂⁺, HCl) and (Cl⁻, CH₃COOH)
 (D) (HCl, Cl⁻) and (CH₃COOH₂⁺, CH₃COOH)

2. Among the following compounds, the strongest acid is

- (A) HC ≡ CH (B) C₆H₆
 (C) C₂H₆ (D) CH₃OH

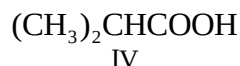
3. Which of the following pairs could undergo a Lewis acid –base reaction ?

- (A) FeCl₃ and Cl₂ (B) CH₃COCl and AlCl₃
 (C) C₂H₅OH and ZnCl₂ (D) All of these

4. Which of the following arrangements gives the correct order of decreasing basicity?

- (A) CH₃O⁻ > CH₃NH₂ > CH₃NH⁻ > CH₃COO⁻
 (B) CH₃COO⁻ > CH₃NH⁻ > CH₃O⁻ > CH₃NH₂
 (C) CH₃NH⁻ > CH₃O⁻ > CH₃NH₂ > CH₃COO⁻
 (D) CH₃NH₂ > CH₃NH⁻ > CH₃COO⁻ > CH₃CO⁻

5. Given the correct order of increasing acidity of the following compound.



- (A) V < II < IV < I < III (B) IV < II < V < III < I
 (C) III < I < IV < II < V (D) V < IV < II < I < III

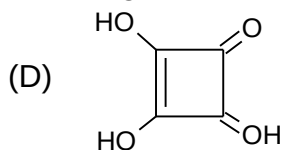
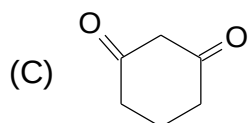
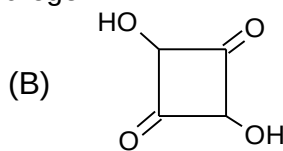
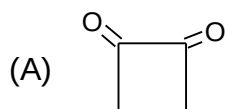
6. Among the following compounds, the strongest base is

- (A) C₆H₅NH₂ (B) p-NO₂C₆H₄NH₂
 (C) m-NO₂C₆H₄NH₂ (D) C₆H₅CH₂NH₂

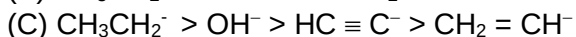
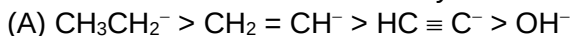
7. The strongest acid among the following is

- (A) CF₃COOH (B) CCl₃COOH
 (C) CBr₃COOH (D) CH₃COOH

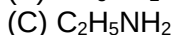
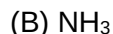
8. Which of the following has the most acidic hydrogen ?



9. Select the correct order of basicity.



10. Which one among the following is the least basic?



LEVEL - IV

A

Matrix-Match Type

1. Match the following column – I and column -II

Column –I	Column –II
(A) Monobasic acid	(p) Cl^-
(B) Soft acid	(q) $\text{H}^+ + \text{OH}^- \rightleftharpoons \text{H}_2\text{O}$
(C) Hard base	(r) H_3PO_2
(D) Neutralisation	(s) Ag^+

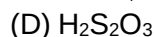
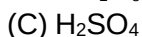
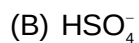
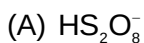
B

Linked Comprehension Type

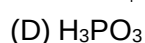
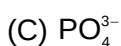
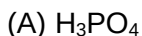
Comprehension - I

Acids are the species capable of donating the proton in a reaction while bases are the species capable of accepting the proton in a reaction, i.e., acids are proton donors (protogenic) and bases are proton acceptors (protophilic).

1. The conjugate acid of $\text{S}_2\text{O}_8^{2-}$ is



2. The conjugate base of H_2PO_4^- is



3. Proton donating species is called
(A) acid (B) base
(C) salt (D) none of these

Comprehension - II

A base is a substance which can furnish a pair of electrons to form a coordinate bond whereas an acid is a substance which can accept a pair of electrons. A Lewis acid is a substance that acts as an electron acceptor or electrophile while a Lewis base is a substance that acts as an electron donor or nucleophile.

4. Which of the following is a Lewis acid?
(A) NH_3 (B) H_2SO_4
(C) H_3O^+ (D) AlCl_3
5. BF_3 is taken as an example of Lewis acid because it behaves
(A) electrophile (B) nucleophile
(C) neutral molecules (D) free radical
6. Which of the following is a Lewis base?
(A) H_2O (B) HCl
(C) NH_4^+ (D) CH_4

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8. Answers to Objective Assignments

LEVEL - I

- | | |
|---------|---------|
| 1. (D) | 2. (A) |
| 3. (C) | 4. (B) |
| 5. (C) | 6. (A) |
| 7. (A) | 8. (A) |
| 9. (A) | 10. (A) |
| 11. (D) | 12. (A) |
| 13. (A) | 14. (A) |
| 15. (B) | 16. (C) |
| 17. (A) | 18. (B) |
| 19. (C) | 20. (C) |
| 21. (B) | 22. (A) |
| 23. (B) | 24. (A) |
| 25. (B) | 26. (A) |
| 27. (D) | 28. (A) |
| 29. (D) | 30. (D) |

LEVEL - II

- | | |
|--------------|------------------|
| 1. (A), (C) | 2. (A), (C), (D) |
| 3. (A, C, D) | 4. (B, C, D) |
| 5. (A, B, C) | |

LEVEL - III

- | | | | |
|--------|---------|--------|--------|
| 1. (D) | 2. (D) | 3. (D) | 4. (C) |
| 5. (B) | 6. (D) | 7. (A) | 8. (D) |
| 9. (A) | 10. (D) | | |

Level - IV

A		B	
Match the following		Write-up	
1.	(A - r)	1.	(A)
	(B - s)	2.	(B)
	(C - p)	3.	(A)
	(D - q)	4.	(D)
		5.	(A)
		6.	(A)